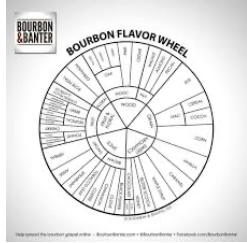


Affective senses

Frédérique de Vignemont

Percept and affect (1)



Affective senses



Bodily sensations



Vision

- Standard model
 - Vision (and audition) does not respond to values.
 - It can only be the basis for emotions.

VISION → EVALUATION

BUT... A comparison

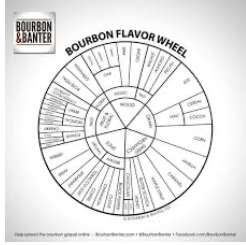


1. Seeing the white wall of your office



2. Seeing a snake

Percept and affect (2)



Affective senses

Bodily sensations



Vision

- A revised model
 - Vision can respond to evolutionary values.
 - Affective phenomenology at the visual level.

EVALUATION  VISION

A dark interest

Focus on negative values

- More experimental studies
- More importantly, it is not a disease to see life through rose-coloured glasses, but it is to look on the black side of things
 - Anxiety disorder in 1 out of 4 people at one point in their life.



A distinction and a question



- **Sensory** unpleasantness
 - Unpleasantness of low-level properties



- **Perceptual** unpleasantness
 - Unpleasantness of high-level properties

BUT is it really the visual experience that feels unpleasant or the anxiety that it triggers?

Unpleasant seeing

A methodological principle



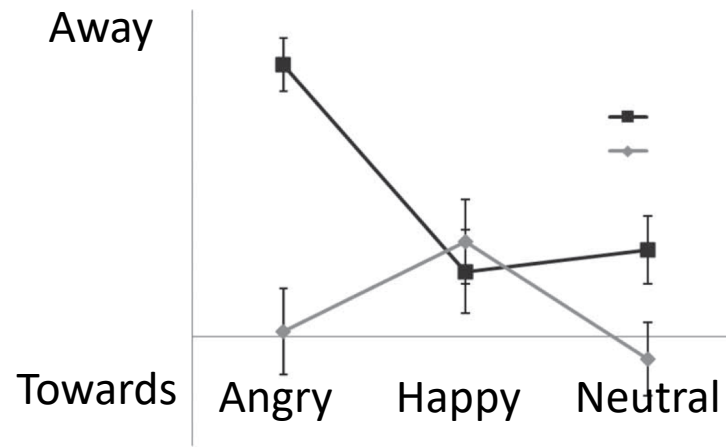
- A parallel with pain
 - How to show that animals feel pain, meaning an unpleasant conscious sensations, instead of nociception?
 - *“Animals can show that they value a putative analgesic or anaesthetic when injured”* (Crumb et al., 2022)

Head-in-the-sand principle

If one has the urge to make a sensation stop, then it can be taken as evidence that the sensation is unpleasant.

Perceptual avoidance

- “Affective” senses: We have the urge to stop unpleasant gustatory or olfactory sensation.
- But also vision and audition
 - Looking away, closing your eyes, looking up rather than down, etc
 - Cf. Schmidt et al. (2012); Ballotta et al., (2025)



A consequentialist view

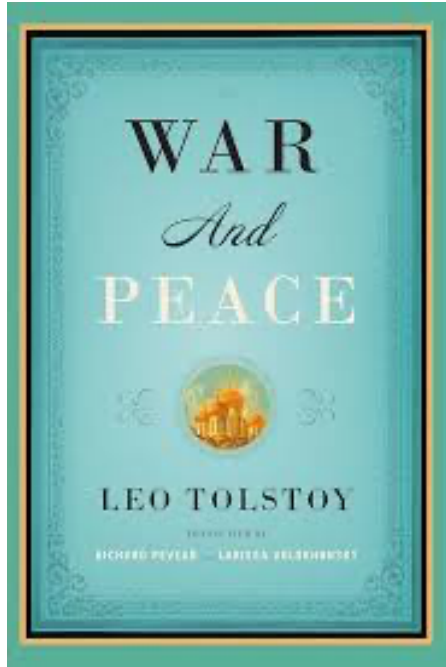
- Iris has a severe toothache. If she takes painkillers, it is only because pain prevents her doing other things (Martinez, 2015).



Revised head-in-the-sand principle

If one has the urge to make a sensation stop, then it can be taken as evidence that the sensation is unpleasant and/or that it has negative consequences.

Tolstoy (1867)



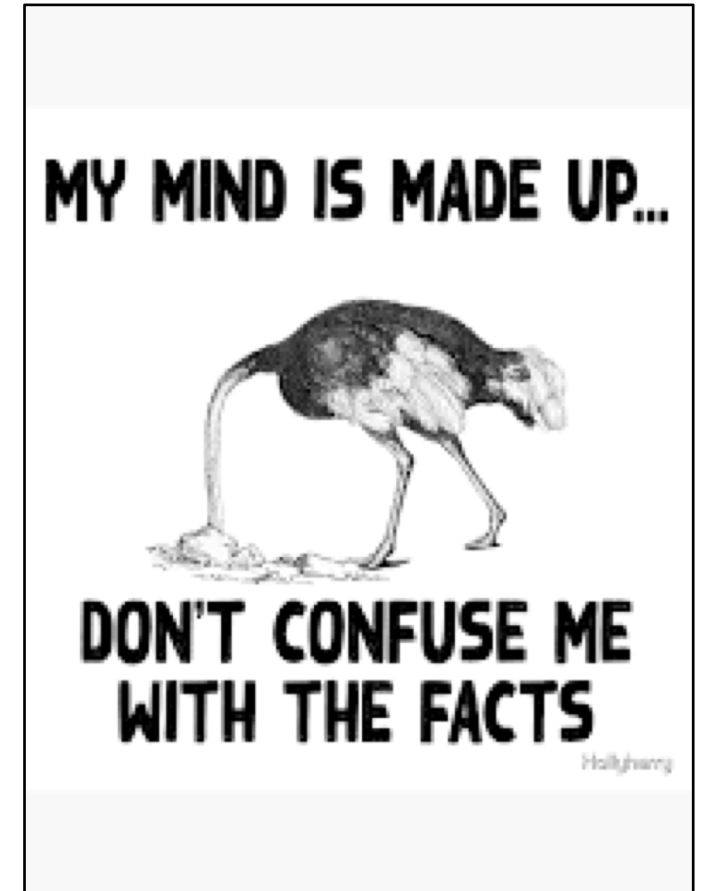
- *“At the approach of danger there are always two voices that speak with equal power in the human soul: one very reasonably tells a man to consider the nature of the danger and the means of escaping it; the other, still more reasonably, says that it is too depressing and painful to think of the danger, since it is not in man’s power to foresee everything and avert the general course of events, and it is therefore better to disregard what is painful till it comes, and to think about what is pleasant.”*

Cognitive and emotional avoidance

- Avoiding information although it is available at no apparent cost, the subject is aware of it, and it would benefit her decision-making (Goldman et al., 2017).
- One may say that you avoid looking down when climbing only because if you were to look down, you would realize how high up you are and that would make you feel anxious.

Too late?

- It would be too easy if for one to stop feeling anxious it sufficed to close one's eyes.
- Emotions typically outlast the perceptual event that triggers them and blocking the perceptual experience does not suffice to block them.
- Distinction between:
 - Conditions of formation: perception
 - Conditions of persistence: memory, imagination, thoughts



A poor strategy



- Blocking the sensory reminder that the object of fear is still present ‘refreshing’ the emotion that it originally triggered.

BUT

- The absence of sensory information about the threat can also be stressful because one is left in the dark about what is happening.
 - Ignorance can reinforce imagination, which in turn can reinforce the emotion.
- Perceptual avoidance is far from being optimal for emotion regulation.

The model of familiarity



A visual experience affectively-loaded

- First, because of a feeling of familiarity
 - Perceptual phenomenology is dual, both sensory and affective (Dokic and Martin, 2015).
- Also because a feeling of danger?
 - Like familiarity, affective response to the broad category of personal significance.

An independent feeling

- Feelings of familiarity consist in self-standing affective experiences that are only integrated with the visual experiences.
- Feeling of déjà vu: familiarity that is not associated with a specific object in the visual experience of the visual scene.
- On this model, visual experiences can be said to be unpleasant in so far as they are integrated with the feeling of danger.
- BUT they are not *intrinsically* unpleasant.
 - Different from affective senses.



A new cake

- However, as we shall now see, the impact of negative value on visual processing is not just a matter of adding a new layer.
- Visual processing itself is modified.
 - It then obeys principles similar to the ones found for pain and fear.
- And like pain and fear, it becomes unpleasant.



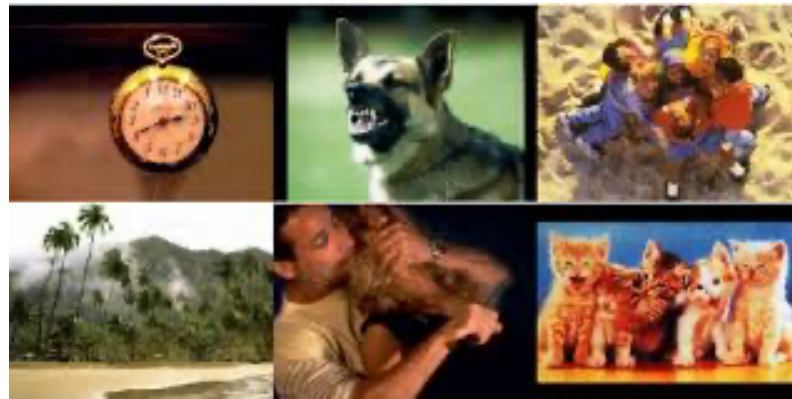
What impact of negative value on vision?

An experimental mess

- Series of experimental findings that show that the visual system does not process threatening objects in the same way as more ‘innocent’ objects.

BUT

- What really drives the effect? The negative value or low-level properties?
 - International Affective Picture System (IAPS)

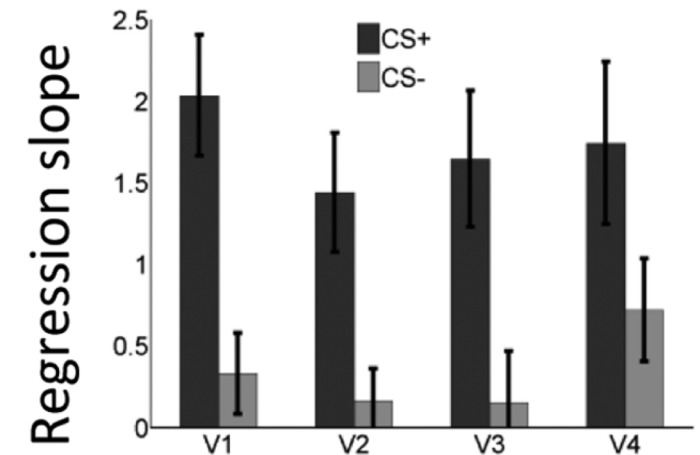


Padmala and Pessoa (2008)

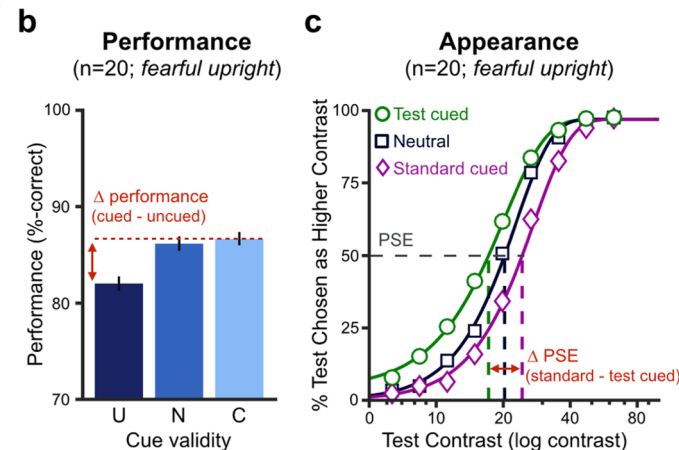
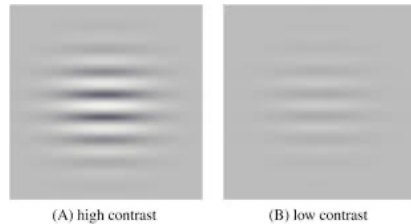
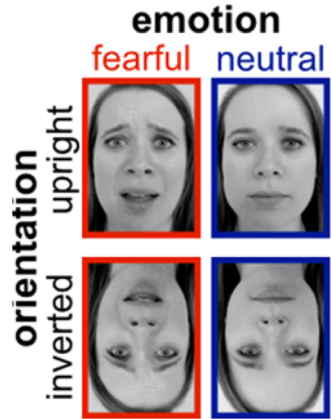
Negative conditioning:



- Results: Participants were faster and more accurate at detecting CS+ than CS-.
- At the neural level, retinotopically organized changes in V1, which were closely correlated with the participants' performance



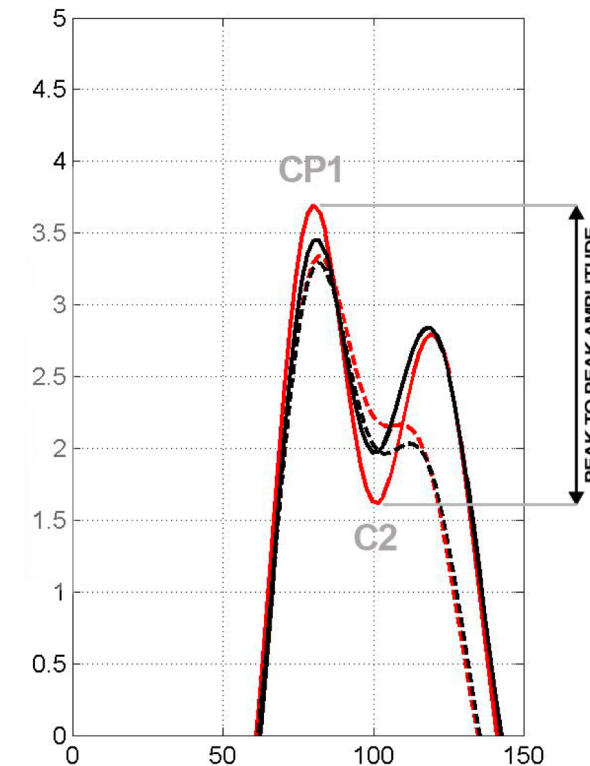
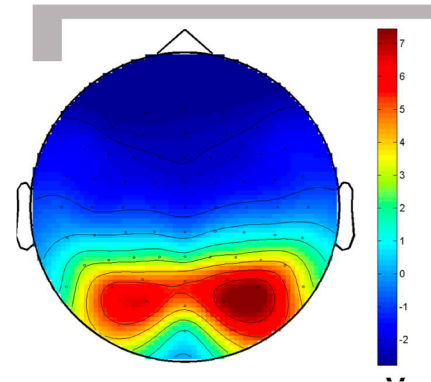
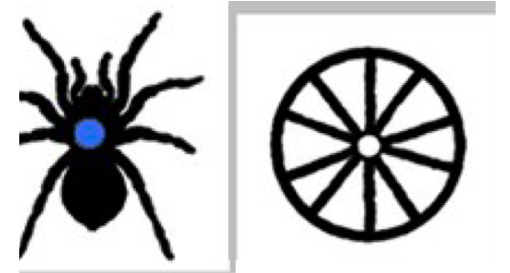
Barbot and Carrasco (2018)



- Participants were asked to report the orientation of the highest contrast Gabor patch.
- Results: After the presentation of a fearful face, improved performance in orientation task.
- Also, significant difference in visual contrast appearance of the Gabor patch.
 - The content of the experience is modified.

Carretié et al. (2022)

- Participants were presented either grey spiders or wheels for only 8ms, immediately followed by a mask for 83ms.
 - They were unaware that they had even seen something, let alone a spider.
- Results: a short-latency sensitivity in EEG to the spider as early as 72ms after the onset.



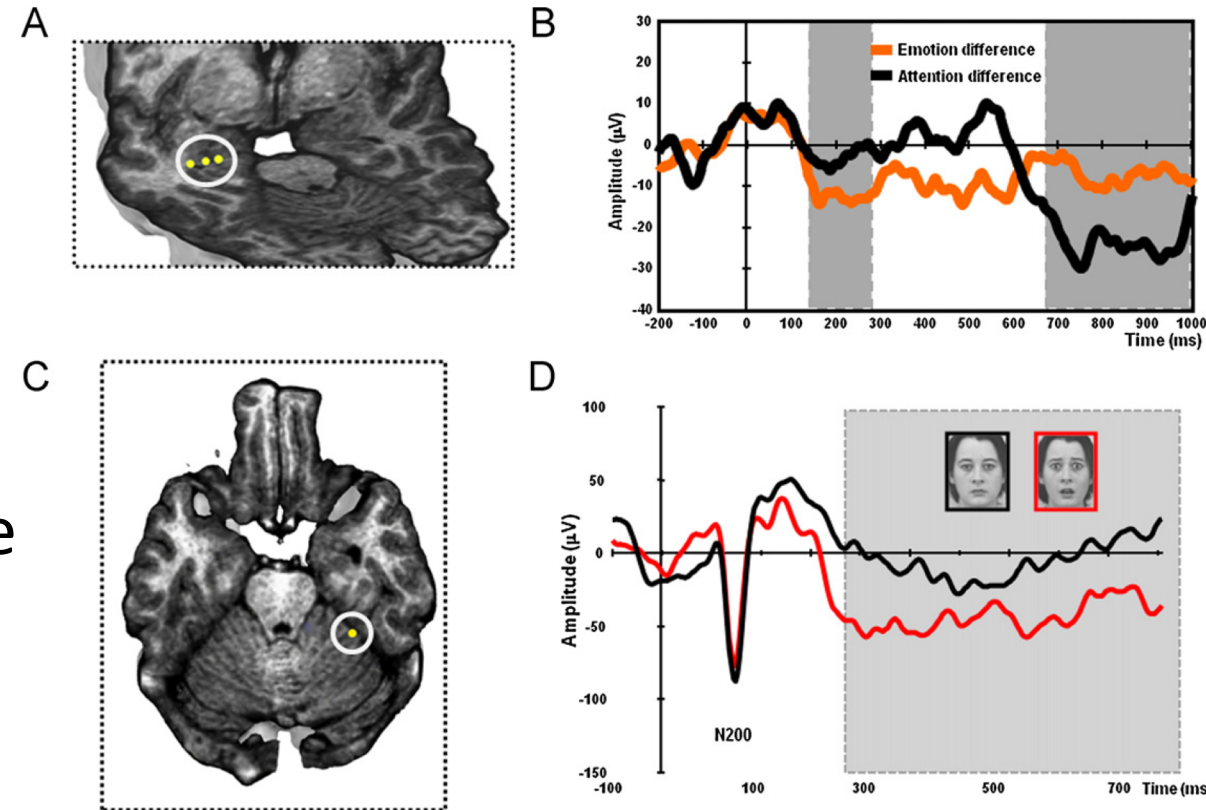
A mere effect of attention?

- There are many things with no evolutionary value that one can actively look at and for which there is no unpleasant feeling.
- You can pay attention to the white wall
 - Because I ask you to do so => endogenous attention
 - Or because something on it has attracted your attention => exogeneous attention.
- Both types of attention then boost your visual processing.
- Is that all there is for threats?



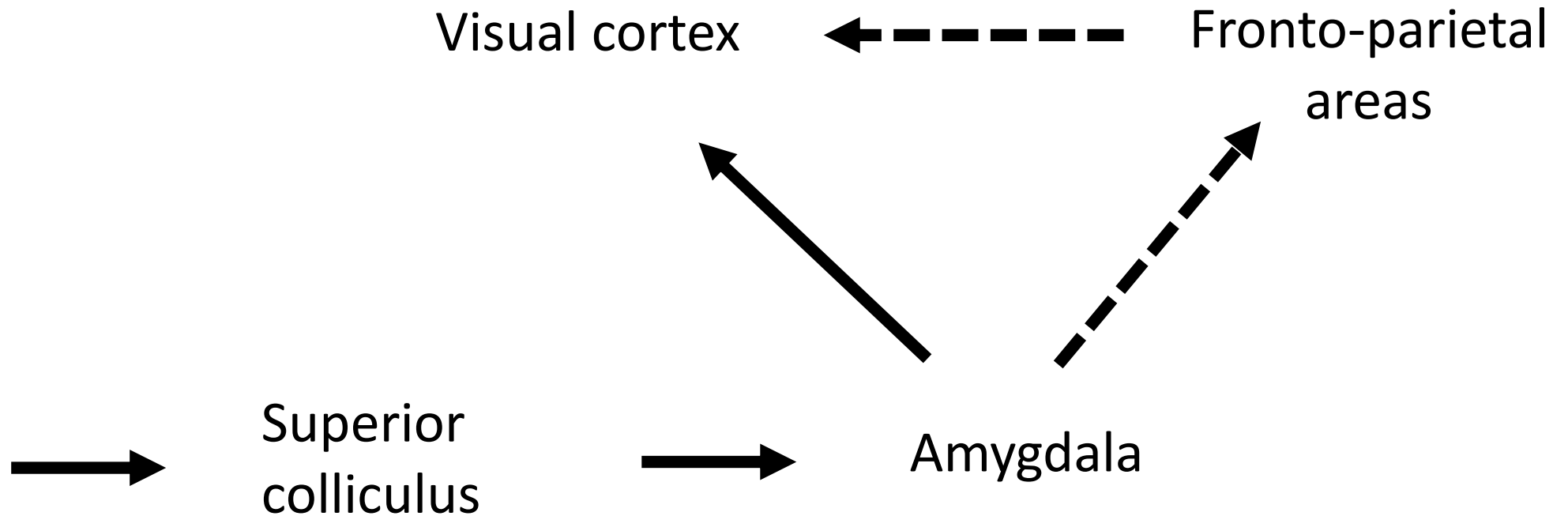
A domain-specific mechanism

- NOT exogenous attention because not captured by low-level visual features but by objects with evolutionary significance.
- NOT endogenous attention because it operates very early on.



Pourtois et al. (2013)

Evaluation → Perception



How specific?

- Affective specialization: the modulation of visual processing cannot be explained by domain-general mechanisms.

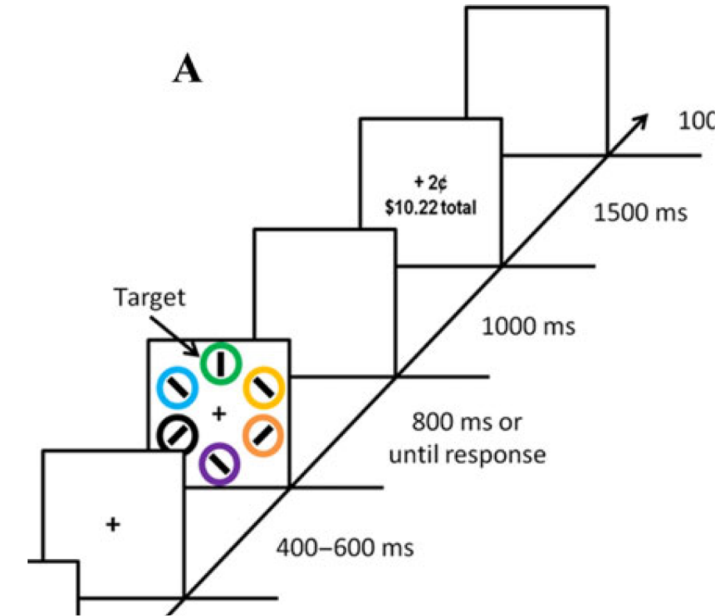
But is it the same for both negative and positive values?

- Because if it is, then it cannot explain unpleasantness.
- No straightforward answer when looking at the experimental literature
 - Positive stimuli are often associated with smaller arousal ratings and less motivational relevance (unless associated with monetary reward).

Reward-association paradigm



- Training phase: monetary rewards associated with colour stimuli.
- Test phase: visual task on the associated stimulus (but no more reward).
- Results: Attentional capture in visual search task by the previously reward-associated distractors (Anderson et al., 2011).
- Enhanced activity in early visual cortex for stimuli that were associated with the reward (Anderson et al., 2014)



But even then...

- No direct comparison with negative stimuli.

“Although electric shock appears to have an especially potent effect on attentional selection, direct comparison to the effects of monetary outcomes on attention is difficult as these sources of feedback differ in many respects. (...) the extent of the similarity in the neural mechanisms underlying the effects of reward and punishment on attention remains to be shown” (Anderson 2016)

The good, the bad, and the... neutral

- General cognitive negativity bias:
 - Stronger response for negative than positive information.
- A matter of timing (Cipriani et al., 2025)
 - Larger amplitude at the early level for negative stimuli
 - Larger amplitude at the later stage for positive stimuli
- Attentional focus (Chaisilprungraung et al., 2024).
 - Attention narrowing after angry face and broadening: after happy face

BUT

- Attention narrowing depends on the motivational value (Gable and Harmon-Jones, 2008)
 - More likely to act on the cake than on a happy face

Beyond empirical data



“sensory systems exemplify (usually quite elegant) solution to specific processing problems” (Akins, 1996)

- Seeing negative and seeing positive stimuli does not give rise to the same processing problems.
 - Thus, different solutions
- 2 types of processing problems
 1. Time-pressure
 2. Precautionary principle

1. Emergency!



“The satisfaction of hunger and sexual appetite can be postponed; not so escape from a dangerous enemy, and all animals, even the biggest and fiercest, have enemies.” (Hediger, 1955)

- Control precedence (Frijda, 1988): priority in the control of attention and action, interrupting other activities, especially if it is intense.
- Not true only of negative emotions, but also of negative perception
 - The earlier the predator is detected and spatially located, the more time is left for you to prepare your response and neutralize it.
 - This explains the prioritization of negative stimuli in the visual system.

A case study: food

- No such time-pressure for even highly motivational positive stimuli, such as food.
 - Hunger strike for 94 days in late 1920, during the Irish War of Independence.
- Even if to be successful, a predator needs to be visually quick to detect the prey before being detected, the prey still has to be faster.
 - The predator can indeed always find another meal, but the prey will not have other opportunities.



2. Better safe than sorry

- An over-zealous visual system
 - What matters most is that we are left unharmed rather than we are correct in our assessment of the situation.
- An asymmetry
 - **False positive**: to visually process as a threat something that is not dangerous => minor cognitive cost
 - **False negative**: to process an actual threat as if it were not one => a possibly fatal cost.



Back to food

- Similar precautionary principle but with opposite asymmetry
 - **False positive**: to process berries as if they were nutritious, while actually they are not => risk of poisoning
 - **False negative** => to process berries as if they were not nutritious, while actually they are => a waste, but no real consequences
- This explains food neophobia in children.



Conclusion

- A first computational step
 - Value modulates visual processing in a distinctive manner
 - The modulation is not the same for the good and the bad.
- A second phenomenological step
 - In virtue of the computational specificities of visual threat processing, visual experiences are intrinsically unpleasant.
- BUT an explanatory gap?
 - From computations to affective phenomenology

An interesting correlation

- At the behavioural and neural levels, the modulation of visual processing is more important in participants with anxiety disorders (e.g. Barbot and Carrasco, 2018)
- At the phenomenological level, it feels worse for them.
 - Whereas it typically feels unpleasant to see a spider on one's bed, it simply feels awful for someone who is arachnophobe.
- Hence, the more visual processing is altered, the more unpleasant it feels.

EVERYTHING
IS FINE.
YOU JUST
NEED TO
LOOK IN
THE RIGHT
PLACE.



— KUH — HEDGEYE